

NVIDIA GeForce RTX 5090

Exponential price, incremental performance



The NVIDIA GeForce RTX 5090 was one of the standout announcements at CES 2025 and NVIDIA pretty much reinforced the fact that in the realm of high-performance computing, the pursuit of groundbreaking graphics technology never ceases. Today, we turn our attention to the pinnacle of visual fidelity and the apple of every gamer's eye, the NVIDIA GeForce RTX 5090. This graphics card, a successor to the already impressive RTX 4090, takes it up a notch with the performance and its pricing. Starting at INR 2,14,000, the RTX 5090 costs as much as 3-5 entry-level PCs put together and to figure out whether it's really worth commanding such a premium, we're putting it through the paces in this review.

Based on NVIDIA Blackwell, the RTX 5090 is the embodiment of NVIDIA's unwavering commitment to generate more frames. With that aim in mind, NVIDIA has really cranked up the frame generation capabilities of this generation. The RTX 5090 aims to expand the bound-



aries of high-end graphics, boasting advancements in neural rendering, AI capabilities, and raw performance. Let's kick off this review of the RTX 5090 by dissecting its specifications.

NVIDIA GEFORCE RTX 5090 SPECIFICATIONS

The NVIDIA GeForce RTX 5090 features 21,760 CUDA cores, 170 4th Generation RT cores, and 680 5th Generation Tensor cores. It has a boost clock of 2.41 GHz and a base clock of 2.01 GHz. The RTX 5090 has 32 GB of GDDR7 memory with a 512-bit interface. It supports 4K resolution at 480 Hz or 8K at 120 Hz with DSC.

Compared to the RTX 4090, the RTX 5090 has more CUDA cores, RT cores, and Tensor cores. It also has a higher boost clock and base clock. The memory capacity is also higher at 32 GB but the RTX 5090 uses GDDR7 memory instead of GDDR6X. The memory interface is also wider on the RTX 5090. And if we compare it to the RTX 3090 then the RTX 5090 has significantly more CUDA cores,

RT cores, and Tensor cores. Not to mention that it also has a higher boost clock and base clock. The process node has not changed and it is the same TSMC 4N node as before except for whatever minor fixes TSMC might have made since its inception. As a result, the additional performance is coming from the generational improvements to the individual core types and by squeezing a lot more transistors. This would also mean that power consumption will spike and thermals would require better management.

We've already seen the design changes that NVIDIA has made to the Founders Edition cards and the AIC partners are weaving their own little magicks to power this megalodon of GPUs.

PERFORMANCE

In 3DMark, the RTX 5090 scores 32,938 in 3DMark Fire Strike Ultra which puts it leagues ahead of the RTX 3090 and the RTX 4090. Time Spy Extreme scored 25,592. The Time Spy Extreme benchmark test, a demanding assessment of a system's graphics processing capabilities, yielded a score of 25,592. This result indicates that the system under test possesses substantial graphical processing power, capable of handling demanding visual workloads, including high-resolution gaming, 3D rendering, and video editing.

In Basemark while using the Vulkan API, the RTX 5090 achieved a score of 52,908, showcasing its prowess in handling demanding graphical workloads. Meanwhile, under the OpenGL API, the card attained a score of 20,167. While notably lower than its Vulkan score, this result still reflects strong performance considering the age and comparative limitations of the OpenGL API. Finally, when tested using the DirectX 12 API, the RTX 5090 delivered its highest score, reaching an impressive 56,957. This result underscores the card's exceptional capabilities when paired with modern gaming titles and applications that leverage the DirectX 12 API.

In Blender which is a great 3D modeling and animation software that's open-source, we saw the RTX 5090 score nearly 7.5K samples per minute in monster, 4K